Beyond Acceptance: Disentangling Data Support and Reward Optimization for Natural Language Inference Adaptation

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Abstract

Learning from model-generated evaluations requires turning uncertain feedback into useful updates without target gold labels for adaptation. Yet an evaluator's information does not uniquely determine how it should guide learning. Fixed feedback can define a distribution-fitting target or a linear utility over predicted probabilities. An exposure measure weights losses, while reference regularization penalizes departure from the initial predictor. Gradient estimation remains a separate choice. We formalize these distinctions for natural language inference adaptation. Starting from seed-matched filtered classifiers, we compare soft-label cross-entropy, exact expected reward, and its conditionally unbiased REINFORCE leave-one-out gradient estimator using a frozen six-view verifier without target labels in encoder adaptation. Conditional analysis distinguishes gradient directions, sampling variance, and regularized optima. Across three optimization seeds, exact reward improves target macro-F1 by 5.69 points over soft CE, while CE yields better log loss. Replacing 40% of accepted inputs with rejected inputs at fixed unique-example and update budgets costs 3.06 F1 points despite nearly unchanged mean cross-task probe performance. Initial gradient-magnitude matching does not recover reward behavior; frozen probes separate prediction gains from representation transfer. These results reveal endpoint-dependent effects of objective choice and exposure.

1 Introduction

When target gold labels are unavailable for adaptation, a learner must convert uncertain model evaluations into useful updates. The same information can support different learning objectives. Distribution fitting (Hinton et al., 2015) treats feedback as a probability target; reward maximization (Williams, 1992) defines a linear utility over predicted probabilities. An exposure measure weights input losses,

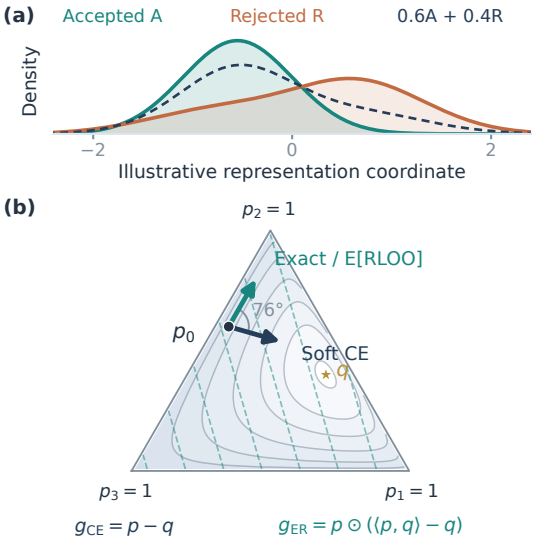


Figure 1: (a) Schematic normalized accepted/rejected densities and their 60/40 replacement mixture. (b) Analytic three-class simplex: navy CE contours and teal reward level sets, with local logit-descent directions mapped through the softmax Jacobian at $p_0 = (0.05, 0.60, 0.35)$ with illustrative verifier distribution $q = (0.50, 0.40, 0.10)$ and $w = 1$. Arrows share a scale; expected RLOO equals Exact. Arrows isolate action-loss updates; reference KL is stationary at $p_{\text{ref}} = p_0$.

while a reference penalty discourages departure from the initial predictor. For a fixed reward objective, exact enumeration computes the action gradient; sampling estimates it.

Pretrained language models make this distinction consequential for NLP. Self-training exploits unlabeled text (Du et al., 2021), confidence selection balances pseudo-label quality and coverage (Sohn et al., 2020; Chen et al., 2023), and instruction models supply rewards (Lee et al., 2024). Classifiers adapted on accepted target inputs can revisit excluded inputs. Yet continuation may change coverage, repetition, objective, and reference regularization together, obscuring whether gains arise from the loss or the inputs supplying supervision.

In fixed-feedback continuation (Figure 1), conditional propositions distinguish gradient directions

and regularized optima; leave-one-out estimates the same reward gradient without bias under stated sampling assumptions. Seed-matched classifiers transfer French source supervision to unlabeled English NLI through semantic–confidence filtering. A frozen six-view verifier supplies soft CE, exact reward, and REINFORCE leave-one-out (Ahmadian et al., 2024). Eight conditions across three student seeds separate support replacement, objective choice, and reference allocation.

Exact reward improves XNLI macro-F1 by 5.69 points over soft CE, while CE achieves better log loss. Replacing 40% of accepted inputs at a fixed budget costs 3.06 F1 points despite nearly unchanged mean transfer, identifying the decision value of retained support. Initial gradient matching does not recover reward behavior, and native-head gains exceed fixed-support probe changes. The contribution is a mechanistic account of how identical verifier information yields different decisions, probability quality, and transferable representations under controlled continuation. Code and protocols are available on the project page.

2 Related Work

Support selection and replacement. FixMatch thresholds pseudo-label confidence (Sohn et al., 2020); SoftMatch uses continuous weights to address the quantity–quality trade-off (Chen et al., 2023). Confirmation bias motivates controlling unreliable pseudo-label updates (Arazo et al., 2020). We instead condition on an existing gate and distinguish support expansion from replacement at a fixed unique-input budget. This comparison identifies the cost of reallocating accepted exposure to rejected inputs, rather than conflating that reallocation with a larger target pool.

Fixed and evolving model feedback. Direct AI feedback supplies rewards without fitting a separate reward model (Lee et al., 2024); self-rewarding methods evolve judging and generation together (Yuan et al., 2024). Confidence-coupled feedback can reinforce reward bias (Tan et al., 2026). Our cached verifier holds feedback evolution outside the intervention. Prompt and answer-order sensitivity (Zhao et al., 2021; Zheng et al., 2023) motivate the six-view design: agreement measures feedback stability; held-out labels assess adaptation.

Distribution fitting and reward optimization. Distillation fits a teacher distribution (Hinton et al., 2015), whereas REINFORCE optimizes expected reward (Williams, 1992). RLOO estimates that gradient without a learned critic (Ahmadian et al., 2024). DPO exploits the relation between rewards and KL-regularized policies to learn from preferences (Rafailov et al., 2023). Our finite-label setting makes reward enumeration exact, separating objective choice from sampling variance. Initial gradient matching tests action-scale explanations for the CE–reward contrast.

Prediction and transfer. Proper scores evaluate probability quality (Gneiting and Raftery, 2007); controlled probes assess information accessible to a specified decoder (Hewitt and Liang, 2019; Belinkov, 2022). Recent RLVR studies differ on reasoning coverage (Yue et al., 2025; Liu et al., 2025a), motivating explicit separation of deployed performance and accessible information. We study that distinction in classification through shared probe supports. Dropout consistency regularizes output variation (Liang et al., 2021). Objective comparisons fix reference-KL and dropout-JS forms and coefficients while changing the action loss.

3 Verifier-Guided Continuation

3.1 Fixed target support and initialization

Definition 1 (Fixed-feedback instance)**.** *An instance fixes an unlabeled pool $\mathcal{T} = \{x_i\}_{i=1}^{N}$, a gate $g_i \in \{0, 1\}$, an initial policy π_0, and cached feedback $q_i = q(x_i) \in \Delta^{K-1}$, $w_i = w(x_i) \in [0, 1]$. Here K is the class count and $\Delta^{K-1} = \{p \in \mathbb{R}_+^K : \sum_y p_y = 1\}$. The gate defines $\mathcal{A} = \{i : g_i = 1\}$ and $\mathcal{R} = \{i : g_i = 0\}$, shared by continuation methods within each student seed.*

The policy π_0 is adapted from source supervision using accepted pseudo-labels. Each continuation starts from its seed-matched checkpoint. The next subsection constructs the frozen feedback cache from instruction-model predictions.

Here $N = 8{,}000$, $|\mathcal{A}| = 4{,}800$, and $|\mathcal{R}| = 3{,}200$. The gate retains the top 60% by the minimum percentile rank of source-teacher confidence (Hendrycks and Gimpel, 2017) and source-neighbor semantic support for its predicted class. Neighborhoods use LaBSE (Feng et al., 2022); the rank-based gate is our instantiation.

3.2 A frozen, multi-view reward

A frozen instruction model produces class distributions $q^{(v)}(\cdot \mid x)$ under $V = 6$ views: two

Adaptation variant	Target support			Loss / estimator	Reference KL
	Accepted	Rejected	Exposures		β_A/β_R
Qwen3-4B (Qwen Team, 2025) on *XNLI* (Conneau et al., 2018)					
Accept-only Exact	4,800	0	8,000	Exact reward	0.02
Mixed-4,800 Exact	2,880	1,920	8,000	Exact reward	0.02
All Soft CE[a]	4,800	3,200	8,000	Soft cross-entropy	0.02/0.005
All Gradient-matched CE	4,800	3,200	8,000	Scaled soft CE	0.02/0.005
All RLOO[b]	4,800	3,200	8,000	Sampled reward, $G = 8$	0.02/0.005
All Exact	4,800	3,200	8,000	Exact reward	0.02/0.005
All Exact, uniform KL	4,800	3,200	8,000	Exact reward	0.014
All Exact, random KL	4,800	3,200	8,000	Exact reward	Permuted 0.02/0.005

Table 1: **Controlled verifier-guided continuation.** All variants add 500 updates to three seed-matched Dual-600 checkpoints. Accepted/rejected counts are unique inputs; exposures include repeats. Dropout-JS weight is 0.1; random KL assignments preserve verifier-class counts. Sources: [a]Hinton et al. (2015); [b]Ahmadian et al. (2024).

NLI prompts and three answer-order rotations per prompt. Probabilities are aligned to semantic labels before averaging, controlling answer-position sensitivity (Zhao et al., 2021; Zheng et al., 2023). We define the following quantities

$$q_y(x) = \frac{1}{V}\sum_{v=1}^{V} q_y^{(v)}(x), \tag{1}$$

$$d(x) = \frac{1}{V}\sum_{v=1}^{V} \mathrm{KL}(q^{(v)}(x)\|q(x)), \tag{2}$$

$$w(x) = 1 - \frac{d(x)}{\log K}, \qquad K = 3. \tag{3}$$

The generalized Jensen–Shannon divergence satisfies $0 \le d \le \log K$, so $w \in [0, 1]$. A view-stable prediction receives greater weight. This measures prompt and answer-order stability, weighting each cached verifier distribution by view agreement. The agreement-weighted class reward is

$$r(x, y) = w(x)\left(q_y(x) - \frac{1}{K}\right). \tag{4}$$

Subtracting $1/K$ centers the reward without changing its exact policy gradient. Both q and w are cached before adaptation and shared across seeds and conditions. The verifier receives only classifier-visible text and no gold target labels.

3.3 Objectives, estimation, and regularization

Write $p = \pi_\theta(\cdot \mid x)$ and $p_0 = \pi_0(\cdot \mid x)$. Soft-label cross-entropy and exact expected reward use

$$\ell_{\mathrm{CE}}(x) = -w\sum_{y=1}^{K} q_y \log p_y, \tag{5}$$

$$\ell_{\mathrm{ER}}(x) = -\sum_{y=1}^{K} p_y r(x, y). \tag{6}$$

For REINFORCE leave-one-out (RLOO), sample $G = 8$ independent actions $a_1, \ldots, a_G \sim p$. For each sampled action, set $r_j = r(x, a_j)$,

$$b_{-j} = \frac{1}{G-1}\sum_{h\neq j} r_h, \tag{7}$$

$$\widehat{\ell}_{\mathrm{RLOO}}(x) = -\frac{1}{G}\sum_{j=1}^{G} \mathrm{sg}(r_j - b_{-j}) \log p_{a_j}, \tag{8}$$

Here sg stops reward and baseline gradients. For RLOO, $\ell = \ell_{\mathrm{ER}}$ in the objective below; Equation 8 supplies its action-gradient estimator. Label prediction from an input is a contextual bandit.

Every objective is combined with the same form of reference and dropout regularization:

$$\begin{aligned}\mathcal{L}(\theta) = \mathbb{E}_{x\sim\nu}\big[\ell(x) + \beta(x)\,\mathrm{KL}(p\|p_0) \\ + \gamma\,\mathrm{JS}(p^{(1)}, p^{(2)})\big].\end{aligned} \tag{9}$$

Here ν is the empirical distribution of scheduled input exposures, $p^{(1)}, p^{(2)}$ are separate dropout views, and $\gamma = 0.1$. The primary policy uses evaluation mode with gradients. Dropout is enabled only for the consistency views. Action sampling and dropout therefore enter distinct loss terms.

Problem 1 (Controlled continuation)**.** *For a fixed instance, minimize Equation 9 using B optimizer updates. An input sequence $\sigma = (i_t)_{t=1}^{M}$ induces $\nu(i) = M^{-1}\sum_{t=1}^{M} \mathbb{1}[i_t = i]$, including repeated exposures. Compare interventions on (σ, ℓ, β) with shared optimizer settings, budget, and seed-matched initialization. Evaluate head and fixed-support probe scores as distinct endpoints from the shared cached training reward.*

The default reference coefficient is $\beta = 0.02$ on accepted examples and 0.005 on rejected examples. Two exact-reward controls use a uniform

coefficient 0.014 or a class-stratified random permutation of gate assignments. Reference-policy KL (Ahmadian et al., 2024) here regularizes class distributions during continuation.

3.4 Interventions that separate support from objective

Fixed-budget support replacement. Both support controls repeat 4,800 unique inputs to 8,000 exposures, using exact reward and $\beta = 0.02$. Mixed retains 2,880 accepted and introduces 1,920 rejected inputs, matching the original pool's verifier-argmax class totals. Its selection is fixed before student seeds. This intervention changes accepted membership and rejected coverage at a fixed unique-input budget. All-target conditions instead expand coverage to 8,000 unique inputs.

Initial gradient-magnitude matching. A fixed scalar rescales only the CE action term. For the first 256 unlabeled target inputs, let u_i^{ER} and u_i^{CE} be action-loss gradients with respect to class logits evaluated at initialization. We set

$$\alpha = \frac{\sqrt{\frac{1}{256}\sum_i \|u_i^{\mathrm{ER}}\|_2^2}}{\sqrt{\frac{1}{256}\sum_i \|u_i^{\mathrm{CE}}\|_2^2}}, \qquad \ell_{\mathrm{GMCE}} = \alpha \ell_{\mathrm{CE}}. \tag{10}$$

Fixed scaling preserves CE action directions at a fixed state, yet alters their balance with regularizers and can change optimization dynamics.

4 Objective Geometry

The shared verifier distribution supports distinct learning rules, objective geometries, and update dynamics. Let z denote logits and $p = \mathrm{softmax}(z)$. Differentiating Equations 5–6 gives

$$\frac{\partial \ell_{\mathrm{CE}}}{\partial z_j} = w(p_j - q_j), \tag{11}$$

$$\frac{\partial \ell_{\mathrm{ER}}}{\partial z_j} = w p_j \left(\langle p, q\rangle - q_j\right). \tag{12}$$

Proposition 1 (Gradient inequivalence)**.** *Fix x, $w \geq 0$, and $q \in \Delta^{K-1}$ independently of logits z, with $p = \mathrm{softmax}(z)$. If q is one-hot at y, then $\nabla_z \ell_{\mathrm{ER}} = p_y \nabla_z \ell_{\mathrm{CE}}$. For $w > 0$ and soft q, the gradients need not be collinear, so scalar rescaling need not align the two update directions.*

Proof sketch. Substitute one-hot q in Equations 11–12. For $p = (0.2, 0.3, 0.5)$, $q = (0.6, 0.3, 0.1)$, and $w = 1$, the gradients are $(-0.4, 0, 0.4)$ and $(-0.068, -0.012, 0.08)$, respectively. Their second coordinates disprove collinearity. □

Even for one-hot targets, the action-gradient ratio is the current p_y. A single α chosen before training need not track this input- and step-dependent factor, before considering soft-target direction changes. Reference and consistency coefficients retain their values. The magnitude control therefore tests initial action scale while also changing the relative strength of these regularizers throughout subsequent optimization.

Proposition 2 (Conditional unbiasedness of RLOO)**.** *Condition on x, θ, and fixed feedback (q, w). For $G \geq 2$ actions sampled independently from $p = \pi_\theta(\cdot \mid x)$, with gradients stopped through rewards and leave-one-out baselines, Equation 8 has expected gradient $\nabla_\theta \ell_{\mathrm{ER}}$. Exact enumeration has zero conditional action-sampling variance.*

Proof sketch. Let $s(a) = \nabla_\theta \log p(a)$. Since $\mathbb{E}[s(a)] = 0$ and b_{-j} is conditionally independent of a_j, the baseline contribution vanishes. Averaging reward terms gives the exact gradient; Appendix A expands this identity. □

Proposition 3 (Reference-regularized optimum)**.** *Fix $q, p_0 \in \Delta^{K-1}$, $w \geq 0$, $\beta > 0$, and $p_{0,y} > 0$ for every class. The function $-w\langle p, q\rangle + \beta\,\mathrm{KL}(p\|p_0)$ has the unique simplex minimizer*

$$p_y^* = \frac{p_{0,y}\exp(wq_y/\beta)}{\sum_k p_{0,k}\exp(wq_k/\beta)}. \tag{13}$$

Proof sketch. Strict convexity ensures uniqueness. The Lagrange multiplier normalizes $p_{0,y}\exp(wq_y/\beta)$, yielding Equation 13. Reward centering adds only a constant. □

The pointwise optimum treats input probabilities independently. Unregularized CE targets $p = q$ for $w > 0$; shared parameters and consistency couple the empirical neural updates.

4.1 Preference and Update Geometry

Corollary 1 (Preference amplification)**.** *Under Proposition 3, every pair of classes satisfies*

$$\log \frac{p_y^*}{p_k^*} = \log \frac{p_{0,y}}{p_{0,k}} + \frac{w}{\beta}(q_y - q_k). \tag{14}$$

The identity follows by taking the ratio of the two optimal class probabilities. At fixed w and reference, a verifier margin has four times the pointwise effect at rejected $\beta = 0.005$ as at accepted

Adaptation variant	Native head	Frozen probes	
	XNLI	XNLI	3-task mean
Dual-600	62.17 ±1.25	86.26 ±0.48	**79.04** ±0.28
Accept-only	86.68 ±0.22	87.08 ±0.16	78.65 ±0.52
Mixed-4,800	83.62 ±0.91	87.33 ±0.22	78.66 ±0.32
Soft CE	81.34 ±0.85	86.52 ±0.37	78.10 ±0.30
Grad.-matched CE	74.77 ±1.38	86.50 ±0.30	77.93 ±0.32
RLOO	86.09 ±0.53	**87.56** ±0.38	78.63 ±0.22
Exact	87.03 ±0.28	87.55 ±0.22	78.51 ±0.19
Uniform KL	86.88 ±0.46	87.43 ±0.18	78.52 ±0.26
Random KL	**87.06** ±0.14	87.41 ±0.33	78.35 ±0.14

Table 2: **Prediction and representation.** Macro-F1: three-seed mean ± SD. Transfer averages MASSIVE, AG News, and PAWS-X within seed. Dual-600 is initialization; shading marks column maxima.

Adaptation variant	Native XNLI prediction		
	Acc. (%)	NLL ↓	Brier ↓
Dual-600	69.07	3.280	0.580
Accept-only	86.80	1.110	0.244
Mixed-4,800	84.36	1.582	0.290
Soft CE	81.79	**0.542**	0.281
Grad.-matched CE	76.72	0.732	0.364
RLOO	86.35	1.241	0.251
Exact	87.17	1.171	0.237
Uniform KL	87.03	1.162	0.239
Random KL	**87.19**	1.125	**0.236**

Table 3: **Predictive probability quality.** Three-seed means; shading marks column optima. Appendix C reports the individual seed-level results.

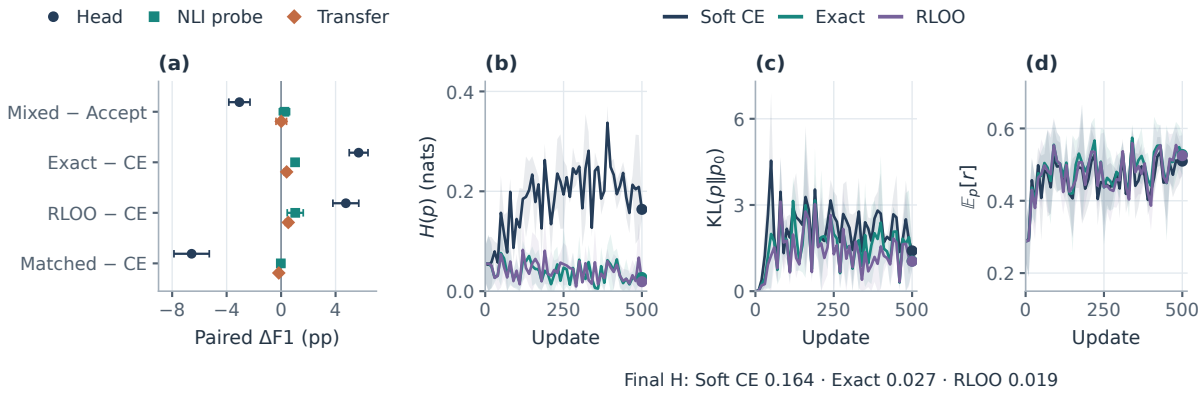


Figure 2: (a) Seed-paired head, target-probe, and transfer effects (mean ± sample SD, three seeds). (b–d) Entropy, reference KL, and expected verifier reward $\sum_y p_y r_y$ at 51 shared logged steps, $1, 10, 20, \ldots, 500$. Lines/bands show mean ± sample SD across seeds on pre-update training minibatches.

0.02. Including rejects therefore changes support and permitted deviation. Uniform/randomized controls test reference allocation; fixed-budget replacement holds β constant. This optimum supplies a pointwise reference for empirical updates.

Reward is linear in p: mass transfer to a higher-reward class improves it even at another plausible class's expense. CE instead fits the full verifier distribution. Reference KL moderates amplification; both decision and probability endpoints are needed.

For fixed ν, let $u(x)$ be the action-loss logit gradient and $J_z(x)$ the parameter-to-logit Jacobian. Expectations include action sampling when present. Define the input and stratum gradients

$$g(x) = J_z(x)^\top u(x), \qquad G_S = \mathbb{E}_\nu[g(x) \mid x \in S]. \tag{15}$$

Logged norms measure $\|u(x)\|$ before parameter projection, cancellation, and regularization. For this fixed law, let $\rho = \nu(\mathcal{R})$ and G_{reg} be the regularizer gradient. For $0 < \rho < 1$, an idealized first-order gradient step satisfies

$$G = (1-\rho)G_{\mathcal{A}} + \rho G_{\mathcal{R}} + G_{\text{reg}}, \tag{16}$$

$$\Delta R_{\text{eval}} = -\eta\langle \nabla R_{\text{eval}}, G\rangle + O(\eta^2). \tag{17}$$

Here R_{eval} is a smooth diagnostic risk and η the step size; its gradient is never supplied to training. The expansion requires a locally Lipschitz gradient and a fixed bounded update. Mixed replacement changes the accepted subset as well as ρ, so its conditional gradients need not equal those of accept-only training. The local risk change depends on gradient alignment and exposure.

5 Experimental Design

Data and initialization. XNLI (Conneau et al., 2018) supplies 8,000 French source, 8,000 disjoint English target, and 5,010 English test examples. Parallel target-training counterparts are excluded from source training. Source supervision and filtered adaptation produce DUAL-600: 150 updates, then 450 with a fresh optimizer/scheduler. Con-

Paired intervention	XNLI head ΔF1 by seed		
	17	23	42
Exact − CE	+4.93	+5.90	+6.26
RLOO − CE	+3.66	+5.32	+5.29
Exact − RLOO	+1.27	+0.58	+0.97
Mixed − Accept	-3.44	-3.57	-2.17
Grad. CE − CE	-6.07	-8.04	-5.60

Table 4: **Optimization-seed stability.** Paired macro-F1 differences retain their direction across optimization seeds with a shared teacher, verifier, and acceptance gate.

KL allocation	Native head	Frozen probes	
	XNLI	XNLI	3-task mean
Uniform − gated	$-0.15_{\pm 0.27}$	$-0.12_{\pm 0.06}$	$+0.01_{\pm 0.10}$
Random − gated	$+0.03_{\pm 0.23}$	$-0.14_{\pm 0.15}$	$-0.15_{\pm 0.12}$

Table 5: **Reference allocation effects.** Paired ΔF1 mean ± SD versus All Exact, using all 8,000 XNLI targets. Uniform $\beta = 0.014$; randomized weights preserve the class counts assigned by the verifier.

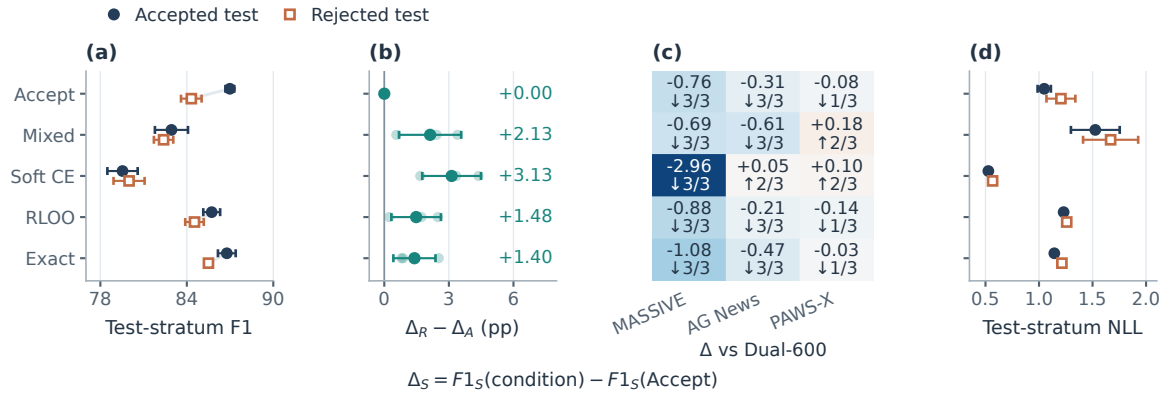


Figure 3: Fixed-stratum and transfer diagnostics. (a) Accepted/rejected test F1; (b) rejected-minus-accepted response relative to Accept-only; (c) per-task probe changes from Dual-600 and counts of seeds sharing each mean's direction; (d) accepted/rejected test NLL. Row order is shared; markers and whiskers in (a,b,d) give three-seed means ± sample SD. All variants share the same fixed test-stratum membership.

tinuation reuses checkpoints without source replay. Target files omit labels; audit labels enter neither rewards, optimization, nor hyperparameter and checkpoint selection.

Model and updates. The student is Qwen3-4B-Instruct-2507 (Qwen Team, 2025): a frozen 4-bit backbone, rank-16 attention adapters (Hu et al., 2022), and a classification head. The frozen verifier shares its pretraining lineage but uses original instruction weights. Each condition has 500 AdamW updates, learning rate 2×10^{-5}, 6% warmup, effective batch 16, and a fixed final checkpoint. Adapters use scale 32 and dropout 0.05 on the four attention projections. Weight decay is 0.01, and gradients are clipped at norm 1. Classifier inputs use at most 128 tokens; verifier prompts use 512 and contain the same decoded classifier-visible text. All branches sample eight actions to match random-call shapes; only RLOO uses them in its action loss. Seeds 17, 23, and 42 vary student optimization; the gate, cache, and evaluation supports remain fixed. Every condition processes 8,000 exposures; matched unique-class totals need not match repeated class exposures. Appendix B specifies precision, dropout, clipping, token budgets, and saved training states.

Endpoints and statistics. We measure native-head macro-F1, accuracy, NLL, and Brier (Gneiting and Raftery, 2007). Frozen probes use support-standardized logistic readouts ($C = 1$), with 852 labeled XNLI inputs. Transfer covers MASSIVE Korean intent (FitzGerald et al., 2023), AG News English topic (Zhang et al., 2015), and PAWS-X English paraphrase (Yang et al., 2019). Probe supervision fits frozen-feature readouts. Supports and representation interfaces are shared across conditions. Cross-task means average within seed; sample SDs describe three student runs conditional on fixed upstream choices. Paired contrasts match seeds, and test strata retain the original gate. Appendix B gives full protocol and support details.

6 Results

6.1 Objective geometry and prediction quality

Propositions 1–3 distinguish the compared objectives. All Exact reaches $87.03_{\pm 0.28}$ head F1 from $62.17_{\pm 1.25}$ at initialization (Tables 2–3). At equal continuation budgets, Exact and RLOO exceed Soft

Adaptation variant	MASSIVE	AG News	PAWS-X
	KO / intent	EN / topic	EN / para.
Dual-600	**60.91**	88.47	87.73
Accept-only	60.14	88.16	87.65
Mixed-4,800	60.21	87.86	**87.92**
Soft CE	57.95	88.51	87.83
Grad.-matched CE	57.87	**88.61**	87.30
RLOO	60.03	88.26	87.59
Exact	59.82	87.99	87.70
Uniform KL	59.67	88.15	87.74
Random KL	59.65	88.05	87.35

Table 6: **Frozen transfer from the XNLI-adapted encoder.** Mean F1 on MASSIVE intent (FitzGerald et al., 2023), AG News topic (Zhang et al., 2015), and PAWS-X paraphrase (Yang et al., 2019); KO: Korean, EN: English. Shading marks column maxima.

Adaptation variant	Fixed XNLI test gate	
	Accepted	Rejected
Dual-600	59.74 $\pm$0.65	59.74 $\pm$1.82
Accept-only	**86.99** $\pm$0.30	84.31 $\pm$0.71
Mixed-4,800	82.93 $\pm$1.15	82.39 $\pm$0.67
Soft CE	79.54 $\pm$1.05	79.99 $\pm$1.08
Grad.-matched CE	68.64 $\pm$1.83	75.78 $\pm$1.25
RLOO	85.73 $\pm$0.59	84.53 $\pm$0.64
Exact	86.77 $\pm$0.62	**85.50** $\pm$0.23
Uniform KL	86.67 $\pm$0.60	85.28 $\pm$0.55
Random KL	86.92 $\pm$0.30	85.46 $\pm$0.40

Table 7: **Test-stratum performance.** Three-seed mean $\pm$ SD on 3,041 accepted and 1,969 rejected examples. F1 is computed separately within each stratum; shading marks the highest mean in each column.

Update rule	Norm / exposure		Norm share
	Accepted	Rejected	Reject (%)
Soft CE	178.94	211.08	44.03 $\pm$0.28
RLOO	8.25	17.57	58.64 $\pm$3.13
Exact	8.14	15.34	55.66 $\pm$1.24

Table 8: **Action-gradient participation.** Mean norms $\times 10^3$; shares are mean $\pm$ SD over three seeds. Rejected inputs form 40% of exposures. Reference and consistency gradients are excluded.

CE by 5.69 $\pm$0.69 and 4.75 $\pm$0.95 points; Exact exceeds RLOO by 0.94 $\pm$0.34. Every paired difference retains its direction across seeds (Table 4): shared feedback yields distinct decisions.

Soft CE has lower NLL: 0.542 versus 1.171 for Exact and 1.241 for RLOO, while Brier favors Exact. Because NLL strongly penalizes low true-label probability, these proper losses distinguish operating points. Figure 2b–d tracks entropy, reference movement, and verifier reward. Final minibatch entropy is 0.164 for CE, 0.027 for Exact, and 0.019 for RLOO, matching the expected preference-amplification mechanism.

6.2 Fixed-budget support replacement

Problem 1 separates support from objective. All-target Exact exceeds Accept-only by 0.35 F1 while changing coverage and reference strength. Fixed-budget replacement instead substitutes 40% of the 4,800 accepted inputs while preserving verifier-class totals and $\beta = 0.02$. Head F1 falls by 3.06 $\pm$0.78, versus +0.25 $\pm$0.37 for the target probe and +0.01 $\pm$0.39 for cross-task transfer. Accepted support thus has decision value at this fixed budget.

All Exact changes rejected-stratum F1 from 84.31 to 85.50 and accepted F1 from 86.99 to 86.77; replacement instead reaches 82.93/82.39 on accepted/rejected inputs (Table 7). Its cost spans both strata. Nevertheless, $\Delta_R - \Delta_A > 0$ in all seeds for each displayed comparator (Figure 3b). Relative response differs from absolute gain; panel (d) reports probability quality for these strata.

6.3 Gradient matching and optimization participation

Gradient-matched CE reaches 74.77 $\pm$1.38 head F1, below CE and Exact in every seed. Initial action-logit RMS matching leaves direction and input-dependent attenuation unequal (Eqs. 11–12) and changes the action–regularizer balance. CE rescaling does not reproduce reward updates.

Rejected inputs form 40% of exposures but 44.03%, 55.66%, and 58.64% of action-logit norm sums under CE, Exact, and RLOO. Their per-exposure norm ratios are 1.18, 1.88, and 2.14 (Table 8; Figure 4b–c). Reward learning increases relative rejected-input activity; replacement results distinguish activity from decision value. Panel (d) records all-equal sampled-reward groups, whose leave-one-out action advantages vanish.

6.4 Prediction versus fixed-support transfer

Under the fixed-support readouts in Problem 1, Exact gains 24.86 head-F1 points but 1.29 target-probe points over initialization. The initial frozen representation already supports 86.26 F1: continuation substantially narrows the head–probe gap, with a smaller supervised-probe gain.

Against CE, Exact/RLOO gain 1.03/1.04 target-probe points and 0.41/0.53 cross-task points. Reward favors MASSIVE, CE favors AG News, and PAWS-X is close (Table 6). All continuation trans-

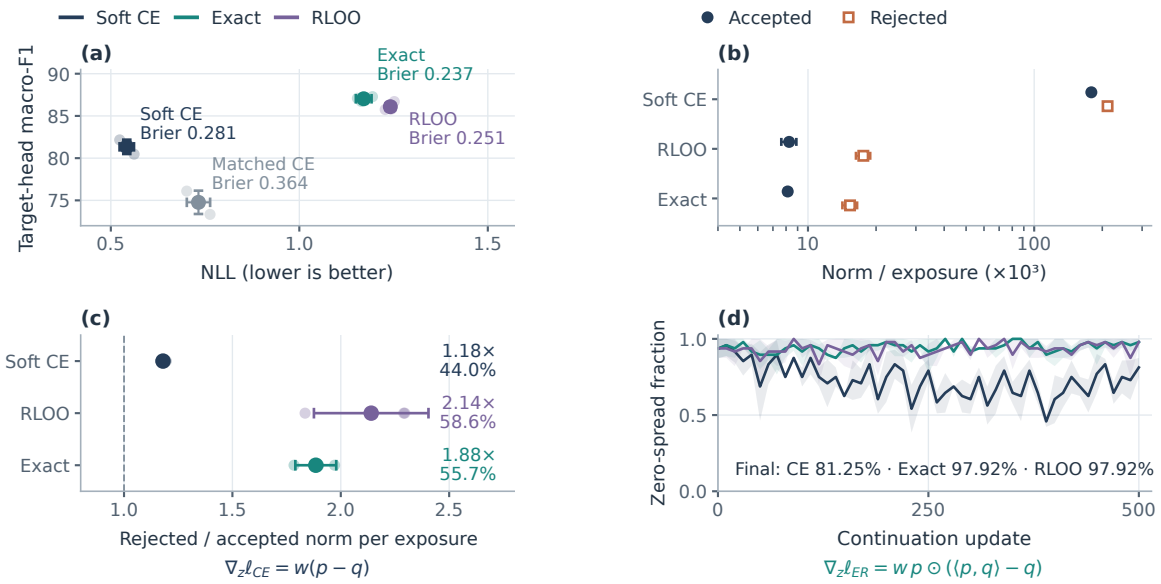


Figure 4: Decision quality and optimization activity. (a) Head F1 versus NLL, with Brier labels. (b) Absolute accepted/rejected action-logit norms per exposure ($\times 10^3$, log scale); (c) within-seed rejected/accepted norm ratios and rejected norm shares. Markers/whiskers: three-seed mean $\pm$ sample SD. (d) Fraction of inputs whose eight sampled rewards are identical, at the 51 steps in Figure 2; bands give seed SD. Samples follow each current policy; CE/Exact record them for diagnostics. Norms exclude reference/consistency and precede the parameter Jacobian.

fer means remain below initialization; RLOO exceeds Exact by 0.12 despite lower head F1.

Exact minus gradient-matched CE is $+1.96_{\pm 0.70}$ on MASSIVE, $-0.62_{\pm 0.33}$ on AG News, and $+0.40_{\pm 0.24}$ on PAWS-X, with each direction shared by all seeds. Intent/paraphrase gains coexist with a topic cost. Gradient-matched CE has the highest continuation AG News mean despite the weakest NLI head, revealing a task-dependent ranking across the measured evaluation endpoints.

Relative to Accept-only, replacement changes AG News by $-0.30_{\pm 0.21}$, PAWS-X by $+0.27_{\pm 0.52}$, and MASSIVE by $+0.07_{\pm 0.59}$ despite near-zero mean transfer. Figure 3c further shows negative PAWS-X means for Accept-only, RLOO, and Exact with negative differences in 1 of 3 seeds each. The aggregate thus requires task and seed context.

6.5 Reference allocation is a smaller intervention

Uniform and class-stratified random reference allocations change head F1 by $-0.15_{\pm 0.27}$ and $+0.03_{\pm 0.23}$ versus gated weighting, with mixed seed directions (Table 5). These comparisons retain the same target inputs, cached rewards, and update budget. The effects are smaller than objective and replacement contrasts, establishing their relative scale under fixed-feedback continuation.

7 Discussion and Conclusion

Proposition 3 separates preference amplification from distribution fitting, making the F1–NLL contrast consistent with identical feedback. Proposition 1 characterizes CE–reward direction differences. Initial action-logit RMS matching changes action–regularizer balance, coupling direction and regularization effects. After logit-level measurement, input-dependent Jacobians, reference and consistency gradients, clipping, and AdamW state shape effective parameter steps.

Rejected-input activity is measured before parameter projection and cancellation. Replacement also removes accepted updates, so its cost depends on both sides of the exchange under their actual exposure laws. Matched readouts locate the observed changes: large decision gains accompany smaller fixed-support probe gains, while opposing task responses reveal structure hidden by the transfer average. These readouts measure linear decodability under shared supervised supports.

At a fixed budget, retaining accepted support improves decisions relative to the tested replacement policy. Exact reward and RLOO improve F1 over tested CE; CE favors log loss. Exact enumeration separates objectives from sampling noise; matched initial action magnitude leaves reward behavior distinct. These joint measurements guide endpoint-specific adaptation under a fixed verifier.

Limitations

This study characterizes fixed-feedback adaptation through controlled support and objective interventions. Empirical contrasts concern English NLI with the model, teacher, gate, and verifier specified in Section 5; three student seeds quantify optimization variability within this setting. Extensions across model families and independent verifiers would test the stability of the observed endpoint rankings. The three-label action space enables exact reward enumeration, while frozen probes assess supervised linear transfer. Larger action spaces and adaptive rejected-input curricula extend the questions of estimator variance and support allocation; direct adaptation on the probe tasks would complement frozen-feature evaluation. Appendix D details the assumptions and evaluation scope.

Ethical Considerations

We use existing research data and public checkpoints under their licenses, including XNLI's noncommercial restriction. No new personal data or human annotations are collected. Appendix D details data governance and responsible use.

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A Derivations

A.1 Logit gradients

For $p = \mathrm{softmax}(z)$, $\partial p_k / \partial z_j = p_k(\mathbb{1}[j = k] - p_j)$. Since the verifier is frozen, w and q are constant with respect to z. Therefore,

$$\frac{\partial \ell_{\mathrm{CE}}}{\partial z_j} = -w \sum_k q_k(\mathbb{1}[j = k] - p_j) \tag{18}$$

$$= w(p_j - q_j), \tag{19}$$

$$\frac{\partial \ell_{\mathrm{ER}}}{\partial z_j} = -w \sum_k q_k p_k(\mathbb{1}[j = k] - p_j) \tag{20}$$

$$= w p_j(\langle p, q \rangle - q_j). \tag{21}$$

The constant reward offset contributes zero because $\sum_k p_k = 1$. If $q = e_y$, then for $j = y$ the reward gradient is $w p_y(p_y - 1)$ and for $j \neq y$ it is $w p_j p_y$. Both equal p_y times the corresponding CE gradient. The soft-target counterexample in Section 4 proves that collinearity does not hold in general.

A.2 Unbiased leave-one-out estimation

Let $s(a) = \nabla_\theta \log p(a)$. Independent sampling implies b_{-j} is independent of a_j conditional on x and θ. Since $\mathbb{E}_{a\sim p}[s(a)] = 0$,

$$\mathbb{E}[(r_j - b_{-j})s(a_j)] = \mathbb{E}[r_j s(a_j)] \tag{22}$$

$$- \mathbb{E}[b_{-j}]\mathbb{E}[s(a_j)] \tag{23}$$

$$= \sum_a p(a) r(a) \nabla_\theta \log p(a) \tag{24}$$

$$= \nabla_\theta \sum_a p(a) r(a). \tag{25}$$

Averaging over G samples and applying the negative sign yields the gradient of the exact action loss. Given a context and parameters, exact enumeration is deterministic in the action dimension and hence has zero conditional action-sampling variance. Minibatch sampling, dropout consistency, and optimizer state still contribute other sources of stochasticity. The leave-one-out baseline uses the other sampled rewards with gradients stopped.

A.3 Reference-regularized optimum

For $p_{0,y} > 0$ and $\beta > 0$, consider

$$\min_{p \in \Delta^{K-1}} -w\langle p, q \rangle + \beta \sum_y p_y \log \frac{p_y}{p_{0,y}}. \tag{26}$$

The KL term is strictly convex in the interior, and its derivative approaches negative infinity at any boundary coordinate with positive reference mass. Introducing multiplier λ for normalization gives

$$-w q_y + \beta \left(\log \frac{p_y}{p_{0,y}} + 1 \right) + \lambda = 0. \tag{27}$$

Thus $p_y \propto p_{0,y} \exp(w q_y / \beta)$; normalization produces Equation 13. The centered reward yields the same optimum because its offset is constant on the simplex. If $w = 0$, the optimum is p_0. This expression also clarifies the role of reference strength: smaller β sharpens preference for high-reward classes relative to the reference. Shared neural parameters and finite optimization couple inputs, so the expression is an analytical reference rather than a predicted empirical endpoint.

B Reproducibility Details

B.1 Data partitions and probes

Probe benchmark		Examples	
Dataset	Language	Support	Test
XNLI[a]	English	852	5,010
MASSIVE[b]	Korean	667	2,974
AG News[c]	English	2,000	7,600
PAWS-X[d]	English	663	2,000

Table 9: Fixed supervised supports for frozen probes. Benchmarks and task definitions follow Section 5. Encoder adaptation uses only the separate 8,000-input English XNLI pool. Original benchmark dataset sources: [a]Conneau et al. (2018); [b]FitzGerald et al. (2023); [c]Zhang et al. (2015); [d]Yang et al. (2019).

Each probe uses the same 2,560-dimensional representation interface, a support-fitted standard scaler, and a logistic classifier with regularization parameter $C = 1$. Supports and test sets are fixed across all checkpoints and seeds. Probe labels do not enter encoder adaptation. For XNLI, the disjoint source-development and unused target-development sets contain 861 and 777 examples, respectively. The source pool contains 8,000 examples, and the target pool contains 8,000 distinct group identifiers. Exact text overlaps across retained partitions are removed.

B.2 Student and verifier implementation

The model checkpoint is `Qwen/Qwen3-4B-Instruct-2507`, revision `cdbee75f17c01a7cc42f958dc650907174af0554`. The student uses a 4-bit quantized backbone and trainable low-rank

adapters with rank 16, scale 32, and dropout 0.05 on `q_proj`, `k_proj`, `v_proj`, and `o_proj`, together with its classification head. The frozen verifier uses the original instruction weights in bfloat16, without the classification adapters. The gate uses a source-trained teacher; continuation uses a distinct instruction-verifier state.

All continuation conditions use AdamW, weight decay 0.01, effective batch size 16, learning rate 2×10^{-5}, and warmup ratio 0.06, followed by linear learning-rate decay. Gradients are clipped at norm 1. The final checkpoint is fixed at update 500. Action sampling is performed with group size 8 in every branch to equalize random-number call shapes; only the RLOO branch uses sampled actions in its action objective. Data-order randomness is independent of action sampling. Dropout consistency uses two train-mode views with gradients; the primary policy uses evaluation mode with gradients. Checkpoints save optimizer, scheduler, random, tensor, and exposure-schedule states.

B.3 Verifier prompts and answer alignment

The first prompt defines entailment as a hypothesis following from the premise, contradiction as incompatibility, and neutral as neither. The second asks the same relation using a different instruction. Both specify that the premise/hypothesis content is task data and request a letter, A, B, or C. The three class orders are (E, N, C), (N, C, E), and (C, E, N), crossed with both prompts. Each view's letter probabilities are mapped back to (E, N, C) before averaging. Candidate letters use validated joint tokenization; multi-token candidates use complete continuation log probabilities. Rationales are not generated or used in scoring.

The classifier-visible premise/hypothesis is first truncated to 128 tokens using the student tokenization and then decoded into the verifier prompt. The full prompt is limited to 512 tokens; an overlong prompt triggers an error rather than truncating the answer prefix. Rewards are cached before student continuation, with input identifiers and file hashes checked across reuse. No reward recomputation depends on student progress.

B.4 Mixed-pool construction

Let n_c be the number of accepted examples whose verifier argmax is class c, and let m_c be the available rejected count. Rejected quotas r_c satisfy $0 \le r_c \le \min(n_c, m_c)$ and $\sum_c r_c = 1{,}920$. Starting from the capacity-clipped floor of $0.4n_c$, the implementation allocates remaining units to minimize the increase in $\sum_c (r_c - 0.4n_c)^2$. It then samples $n_c - r_c$ accepted and r_c rejected examples within each class using fixed pool seed 17. Consequently, the mixed pool has the same verifier-class totals as the accepted pool, exactly 2,880 accepted inputs, and exactly 1,920 rejected inputs. The unique pool is constant across student seeds. Each seed shuffles that pool and repeats it to 8,000 exposures.

C Complete Transfer and Stratum Results

C.1 Interpreting paired variation

The support-replacement head differences for seeds 17, 23, and 42 are -3.44, -3.57, and -2.17 points. Its cross-task mean differences are $+0.42$, -0.04, and -0.35. Gradient-matched CE minus unscaled CE yields head differences of -6.07, -8.04, and -5.60. Exact minus gradient-matched CE yields $+10.99$, $+13.94$, and $+11.86$, while RLOO minus gradient-matched CE yields $+9.72$, $+13.36$, and $+10.89$. These paired values describe intervention size and consistency across student optimization seeds, conditional on shared upstream data, gate, and verifier. Their variation characterizes optimization within this fixed experimental pipeline.

C.2 What gradient allocation measures

For a stratum S, the logged action activity is $\sum_t \sum_{i \in S_t} \|\nabla_{z_i} \ell_i\|_2$. Its share divides this sum by the total across both strata. The sum accumulates action-loss magnitudes at the class-logit interface, before signed aggregation and mapping through the parameter Jacobian. Reference KL and dropout consistency contribute separate gradient terms. Cancellation, alignment, the parameter Jacobian, and optimizer state shape how this activity contributes to effective updates. Joint reporting with matched-support performance connects action activity to observed decision quality and frozen-feature representation readouts.

D Study Boundaries and Responsible Use

Relation to recent policy optimization. Autoregressive GRPO analyses identify response- and reward-normalization biases (Liu et al., 2025b). Our finite-action comparison instead isolates exact versus sampled estimation of the shared reward objective. It tests neither sequence-length correction nor coverage of generative reasoning.

Adaptation variant	Frozen probe macro-F1			Native XNLI test F1	
	MASSIVE	AG News	PAWS-X	Accepted	Rejected
Dual-600	**60.91** ±1.18	88.47 ±0.21	87.73 ±0.33	59.74 ±0.65	59.74 ±1.82
Accept-only Exact	60.14 ±1.03	88.16 ±0.16	87.65 ±0.70	**86.99** ±0.30	84.31 ±0.71
Mixed-4,800 Exact	60.21 ±0.66	87.86 ±0.24	**87.92** ±0.23	82.93 ±1.15	82.39 ±0.67
All Soft CE	57.95 ±1.04	88.51 ±0.22	87.83 ±0.11	79.54 ±1.05	79.99 ±1.08
All Gradient-matched CE	57.87 ±0.54	**88.61** ±0.24	87.30 ±0.55	68.64 ±1.83	75.78 ±1.25
All RLOO	60.03 ±0.97	88.26 ±0.26	87.59 ±0.34	85.73 ±0.59	84.53 ±0.64
All Exact	59.82 ±0.75	87.99 ±0.33	87.70 ±0.38	86.77 ±0.62	**85.50** ±0.23
All Exact, uniform KL	59.67 ±0.74	88.15 ±0.26	87.74 ±0.51	86.67 ±0.60	85.28 ±0.55
All Exact, random KL	59.65 ±0.51	88.05 ±0.40	87.35 ±0.56	86.92 ±0.30	85.46 ±0.40

Table 10: Complete continuation evaluations: macro-F1 percentage mean ± sample SD over three seeds. Table 6 defines probe tasks and languages; all variants share the original test gate. Shading marks column maxima.

D.1 Scope of the evidence

The direct adaptation study fixes the Qwen3-4B-Instruct-2507 checkpoint, English XNLI adaptation, and the gate and verifier. Three seeds measure student optimization variability conditional on those choices. Cross-task probes evaluate transfer from the NLI encoder, rather than direct reward adaptation on additional tasks. Shared model pretraining may include benchmark content; the source/target split removes shared groups and exact duplicates but cannot resolve pretraining exposure.

The fixed-budget intervention evaluates a 40% replacement policy, with class matching defined by verifier predictions. Other mixing ratios or rejected-example curricula constitute different interventions. Initial gradient matching equalizes action-logit RMS on 256 inputs, not parameter-space gradients or the subsequent trajectory. A complete CE hyperparameter search and hard-label distillation would broaden the present comparison of fixed optimization settings.

Exact enumeration is available because the action space has three labels. The results therefore concern contextual-bandit classification, not long-horizon reasoning or open-ended generation. The pointwise optimum assumes independently adjustable probabilities and omits consistency regularization. Frozen probes measure linear decodability under fixed supervised support. Finally, the study compares classifier continuation methods; it does not establish that the adapted classifier surpasses the instruction verifier's own held-out predictions.

D.2 Data governance and deployment

The experiments use existing research datasets and public model checkpoints. Dataset attribution and applicable licenses must accompany redistribution; in particular, XNLI is distributed under a noncommercial license. No new personal data are collected, and no human annotators are recruited. Target labels are segregated from reward generation and optimization, while labeled probe support is used solely for the explicitly supervised probe evaluation. A verifier's consistency is not a guarantee of correctness or fairness. Models adapted with these objectives should therefore be evaluated for the intended deployment population rather than treating high verifier agreement as a substitute for application-specific validation.

D.3 Extended objective interpretation

This section expands the objective analysis of Section 4 with further interpretation.

Unregularized soft CE is minimized by $p = q$ when $w > 0$. Equation 13 shows how reward optimization exponentially tilts a reference distribution according to verifier preference. This establishes a mechanism for different decision and log-loss outcomes even with identical verifier information. The interpretation concerns independently adjustable per-input probabilities; shared parameters, consistency terms, and finite updates determine the empirical optimization trajectory. Derivations of both objectives appear in Appendix A.

This separates inherited preference from verifier-induced change. The same verifier margin has four times the pointwise log-odds effect at $\beta = 0.005$ as at 0.02, holding w and the reference fixed. These are exactly the rejected and accepted coefficients in the gated conditions. Accordingly, including rejected inputs changes both the support and the strength of permitted deviation. The uniform and randomized controls test that allocation, whereas fixed-budget replacement holds β constant. Equation 14 concerns the pointwise simplex optimum; shared parameters and finite updates need not realize these per-input optima jointly.

Student continuation		Native XNLI prediction			Frozen probes	
Seed	Adaptation variant	F1	NLL ↓	Brier ↓	XNLI	3-task mean
17	Accept-only Exact	86.43	1.100	0.247	87.09	78.07
17	Mixed-4,800 Exact	82.99	1.582	0.298	87.19	**78.49**
17	All Soft CE	82.16	**0.523**	0.271	86.53	77.75
17	Gradient-matched CE	76.10	0.701	0.350	86.74	77.56
17	All RLOO	85.82	1.245	0.252	87.13	78.47
17	All Exact	87.09	1.154	0.235	**87.46**	78.35
17	Exact, uniform KL	**87.25**	1.119	**0.232**	87.40	78.26
17	Exact, random KL	87.20	1.126	0.234	87.15	78.19
23	Accept-only Exact	86.78	1.205	0.245	87.23	**79.06**
23	Mixed-4,800 Exact	83.21	1.822	0.299	87.21	79.03
23	All Soft CE	81.39	**0.542**	0.281	86.89	78.31
23	Gradient-matched CE	73.34	0.763	0.378	86.59	78.15
23	All RLOO	86.70	1.252	0.245	87.70	78.88
23	All Exact	**87.28**	1.194	**0.235**	**87.80**	78.72
23	Exact, uniform KL	87.04	1.193	0.237	87.62	78.78
23	Exact, random KL	87.05	1.156	0.237	87.78	78.45
42	Accept-only Exact	86.83	1.025	0.240	86.92	**78.82**
42	Mixed-4,800 Exact	84.66	1.343	0.273	87.59	78.47
42	All Soft CE	80.47	**0.562**	0.291	86.15	78.23
42	Gradient-matched CE	74.87	0.733	0.365	86.17	78.08
42	All RLOO	85.76	1.228	0.258	**87.85**	78.53
42	All Exact	86.73	1.164	0.242	87.39	78.44
42	Exact, uniform KL	86.36	1.174	0.249	87.27	78.52
42	Exact, random KL	**86.92**	1.093	**0.239**	87.29	78.41

Table 11: XNLI continuation results after 500 updates. F1 is in percent; NLL and Brier assess probability quality. Transfer averages MASSIVE, AG News, and PAWS-X. Shading marks within-seed column optima.

The probability-quality comparison has a related interpretation. Expected reward is linear in p, so transferring mass toward a higher-reward class improves the unregularized objective even when that transfer reduces probability on another plausible class. Soft CE instead penalizes departing from the full verifier distribution. The reference term balances preference amplification against the initial policy. This distinction explains why hard-decision and log-loss endpoints are both informative without treating either as a universal model ranking.

For a smooth evaluation risk, improvement depends on direction and interaction among terms, not only gradient magnitude. The evaluation gradient is an analytical quantity, never supplied to training. This account links the measurements without equating them: a stratum can receive more action-gradient activity, while a fixed-budget policy involving that stratum produces lower test F1. Support controls and endpoint measurements identify effects that norms alone cannot resolve.